Stochastic Dynamics and Derivative Pricing

Presentation for the course of Quantitative Financial Modelling

Leonardo Satoru Alosi **Sahar JafariDoust** **Giuseppe Pecorella**
Amina Migdanova **Sebastiano Rocchi**

Master's degree in finance and insurance: financial risk and data analysis

June 2026

Agenda

1. Ito process definition
2. Stochastic Foundations
3. Black–Scholes PDE derivation
4. Merton Extension
5. Computational Prototype
6. Ornstein–Uhlenbeck process
7. Results Interpretation

Ito Process and Stochastic Dynamics

An Ito process is defined as the process that solves the following stochastic differential equation **SDE**:

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t \tag{1}$$

where:

- $\mu(t, X_t)$ = drift term
- $\sigma(t, X_t)$ = volatility term
- W_t = standard Brownian motion

Key intuition:

- Drift represents deterministic evolution
- Brownian motion generates randomness
- We can model financial asset prices through stochastic differential equations !

Why Stochastic Models?

Asset prices evolve with both predictable and random components.

- The predictable component is represented by the drift.
- The random component is represented by Brownian motion.
- This allows us to model uncertainty in financial markets.

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{2}$$

- $\mu S_t dt$: expected growth component.
- $\sigma S_t dW_t$: random market shock.
- Risk-free bond evolves like:

$$dB_t = rB_t dt \tag{3}$$

Derivation of Ito's Lemma

For a smooth function $f(t, X_t)$:

$$df(t, X_t) = f_t\, dt + f_x\, dX_t + \frac{1}{2} f_{xx} (dX_t)^2 \tag{4}$$

Substituting $dX_t = \mu\, dt + \sigma\, dW_t$ and using the multiplication rules:

$$dt^2 = 0, \qquad dt\, dW_t = 0, \qquad (dX_t)^2 = \sigma^2\, dt \tag{5}$$

we obtain:

Itô's Lemma one dimensional case

$$df(t, X_t) = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x\, dW_t \tag{6}$$

Black–Scholes Market Assumptions

The Black-Scholes Framework: Key Assumptions

- **Frictionless markets:** there are no transaction costs, taxes, or bid–ask spreads.
- **Continuous trading:** investors can continuously rebalance their portfolios.
- **Constant parameters:** the risk-free interest rate r and the volatility σ are known and remain constant over time.
- **No arbitrage:** any self-financing riskless portfolio must earn exactly the risk-free rate r.
- **No dividends:** the standard model assumes that the underlying asset does not pay dividends.
- **Short selling allowed:** investors may short-sell the underlying asset without restrictions.

Self-Financing and Delta Hedging

To derive the Black–Scholes PDE, we construct a hedged portfolio using the derivative security and δ shares of the underlying asset.

$$V_t = v(S_t, t) + \delta S_t \tag{7}$$

Choosing the hedge ratio

$$\delta = -\frac{\partial v}{\partial S_t} \tag{8}$$

It allow us to eliminates the stochastic component of the portfolio dynamics.

- The portfolio becomes instantaneously riskless.
- By no-arbitrage, it must earn the risk-free rate r.
- This leads directly to the Black–Scholes PDE.

Black–Scholes PDE Derivation

Starting from the derivative value:

$$v = v(S_t, t) \tag{9}$$

applying Itô's Lemma gives:

$$dv = \left(\frac{\partial v}{\partial t} + \mu S_t \frac{\partial v}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} \right) dt + \sigma S_t \frac{\partial v}{\partial S_t} dW_t \tag{10}$$

- A hedged portfolio is constructed using the option and the underlying asset.
- Choosing the hedge ratio eliminates the stochastic dW_t term.
- The riskless portfolio must earn the risk-free rate r.

Black–Scholes PDE Derivation (cont'd)

For the hedged portfolio $V_t = v(S_t, t) + \delta S_t$, the dynamics are:

$$dV_t = \left(\frac{\partial v}{\partial t} + \mu S_t \frac{\partial v}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} + \delta \mu S_t \right) dt + \left(\sigma S_t \frac{\partial v}{\partial S_t} + \delta \sigma S_t \right) dW_t \tag{11}$$

We set δ the hedge ratio, to make stochastic term vanishes and the portfolio is riskless.

By no-arbitrage:

$$dV_t = rV_t dt \tag{12}$$

Substituting and rearranging gives:

Black–Scholes PDE

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} + rS_t \frac{\partial v}{\partial S_t} - rv = 0 \tag{13}$$

Merton Extension with Dividends

The standard Black–Scholes model assumes that the underlying asset pays no dividends.
Merton extends the framework to assets paying a continuous dividend yield q.
Under the risk-neutral measure, the stock dynamics become:

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t^Q \tag{14}$$

Therefore, the Black–Scholes PDE becomes:

Black–Scholes PDE with dividends

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 v}{\partial S_t^2} + (r - q)S_t \frac{\partial v}{\partial S_t} - rv = 0 \tag{15}$$

- The only structural change is $r \to r - q$.

Computational Prototype

The code starts implementing:

- Simulation of a Geometric Brownian Motion (GBM) path for the stock price S_t.

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right] \tag{16}$$

- Numerical verification of Itô's Lemma using the transformation $X_t = S_t^2$.

Key idea:
The transformed process preserves the same Brownian source of randomness, but its drift and volatility are modified by Itô's Lemma.

Computational Prototype

Applying Itô's Lemma with $X_t = S_t^2$ gives:

$$dX_t = 2S_t\, dS_t + (dS_t)^2 \tag{17}$$

Substituting $dS_t = \mu S_t\, dt + \sigma S_t\, dW_t$ and using $(dW_t)^2 = dt$:

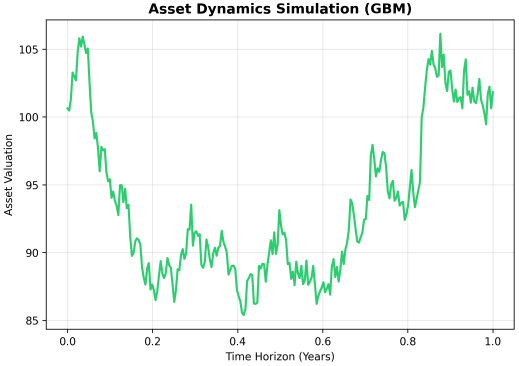
$$2S_t\, dS_t = 2\mu X_t\, dt + 2\sigma X_t\, dW_t \tag{18}$$

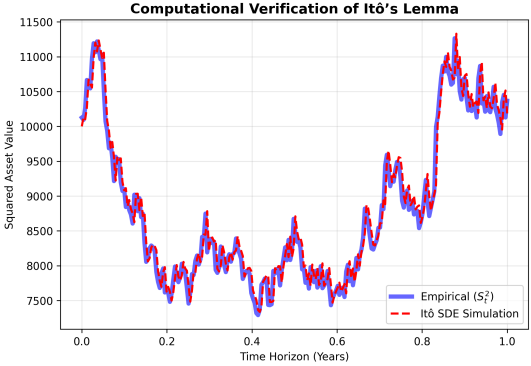
$$(dS_t)^2 = \sigma^2 S_t^2\, dt = \sigma^2 X_t\, dt \tag{19}$$

Summing the two contributions:

$$dX_t = (2\mu + \sigma^2)\, X_t\, dt + 2\sigma\, X_t\, dW_t \tag{20}$$

GBM Simulation and Itô Verification





The simulated GBM path represents one possible evolution of the asset price, while the Itô verification compares the direct squared process S_t^2 with the Itô-derived SDE.

From GBM to Mean-Reverting Dynamics

Ornstein–Uhlenbeck Process

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t \tag{21}$$

where:

- κ is the speed of mean reversion
- θ is the long-run mean level

Key Properties:

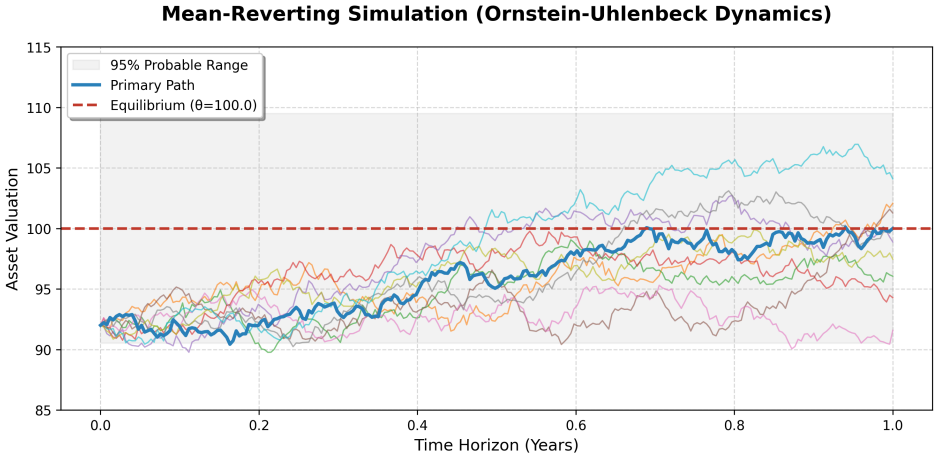
- **Mean:**

$$\mathbb{E}[X_t] = \theta + (X_0 - \theta)e^{-\kappa t} \quad \longrightarrow \quad \lim_{t\to\infty} \mathbb{E}[X_t] = \theta$$

- **Variance:**

$$\mathrm{Var}(X_t) = \frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa t}\right) \quad \longrightarrow \quad \lim_{t\to\infty} \mathrm{Var}(X_t) = \frac{\sigma^2}{2\kappa}$$

Ornstein–Uhlenbeck Simulation



The OU paths fluctuate randomly but tend to revert toward the long-term mean θ.

Conclusion

- Itô's Lemma provides the stochastic chain rule for Brownian-driven processes.
- The Black–Scholes PDE follows from Itô's Lemma, delta hedging, and no-arbitrage.
- The Merton extension incorporates continuous dividends by replacing r with $r - q$.
- The Python prototype illustrates the theory through GBM, Itô verification, and OU simulations.

Bibliography

- Bianchi, S. (2026). *Course Material for Quantitative Financial Modeling*. Course material.
- Black, F. and M. Scholes (1973). "The Pricing of Options and Corporate Liabilities". In: *Journal of Political Economy* 81.3, pp. 637–654.
- Hull, J. C. (2022). *Options, Futures, and Other Derivatives*. 11th ed. Pearson.
- Merton, R. C. (1973). "Theory of Rational Option Pricing". In: *Bell Journal of Economics and Management Science* 4.1, pp. 141–183.
- Shreve, S. E. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer.